Gotzman_Kyle_TermProject_FinalPaper

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DSC640-T302 Data Presentation & Visualization

Term Project: Final Paper

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0.1 The Affordability Crisis in American Childcare: Final Project Paper

0.1.1 Summary of Analysis

This project analyzes the National Database of Childcare Prices (NDCP) from the U.S. Department of Labor. The dataset spans 2008 through 2018 and includes roughly 34,500 county-year records across 3,143 counties. Key fields are weekly median childcare prices, median household income (MHI), and female labor force participation rates.

The main metric is the affordability ratio: annual infant center-based care cost as a percentage of MHI. I compared this ratio to the 7% benchmark used in childcare policy research, then examined state-level variation, decade-long trends, and the relationship between burden and female labor force participation among mothers with children under six.

0.1.2 Findings

The median county-year affordability ratio is about 16%, more than double the 7% threshold. Over 99% of valid county-year observations exceed 7%. Even the lowest-burden states, such as Mississippi near 10%, fail the benchmark.

Geographic gaps are real. Massachusetts and the District of Columbia exceed 23% on a median-county basis. Still, no state clears the affordability line. From 2008 to 2018, childcare prices outpaced income growth, pushing the national median burden upward.

The labor force link is weak. Pearson correlation between burden and female LFP is roughly -0.06. That is a slight negative association, not evidence that cost alone drives parents out of work. Local industry, informal care, and cultural norms likely matter more.

0.1.3 Assumptions

Annual cost assumes 52 weeks at the weekly median, which overstates part-year or part-time use. The 7% threshold is a policy heuristic, not a legal standard. The analysis uses county medians, not household-level data. Imputed prices in sparse counties may add error. Rows with missing MHI or price were excluded from ratio calculations via `dropna()`, reducing the analytic sample to about 23,600 observations.

0.1.4 Items That Still Need Clarification

The dataset ends in 2018, so post-pandemic staffing shortages and inflation are not captured. A full sensitivity run excluding imputed records would strengthen the headline claims. The labor force analysis would benefit from controls for poverty and urban/rural status. Center-based infant care is only one arrangement; family home and informal care may tell a different story.

0.1.5 Direction of Story / Plan of Attack

The core message: childcare is unaffordable in nearly every county studied, the problem worsened over the decade, and geography does not excuse any region. The affordability ratio is the throughline because it turns complex price and income data into one number a legislator can read in seconds. The three mediums move from that headline into geographic context, trend evidence, and cautious workforce discussion before closing on policy levers: direct subsidies, refundable tax credits, and childcare workforce investment.

0.1.6 Target Audience

Primary audience: state and federal legislators and policy staff working on early childhood education, workforce development, and family economic security. Secondary audience: advocacy groups and journalists who need credible national evidence for briefings. These readers have limited time, so clarity and source transparency matter more than statistical depth.

0.1.7 Mediums Included and Why

Static infographic: One-page leave-behind for a 60-second scan in a briefing packet. Headline statistic, map, and trend with no login required.

Interactive dashboard: Filter-driven Plotly draft for staff who need home-state numbers, peer comparisons, and trend context for memos.

Slide deck: Six slides for committee testimony, one idea per slide, sized for projection. Closes with numbered policy recommendations.

Each format matches a different step in the policymaking workflow: quick impression, self-directed exploration, and formal presentation.

0.1.8 Design Decisions

Navy and steel blue headers signal authority; tomato and orange highlight threshold breaches only. Choropleth maps use a reversed red-yellow-green scale anchored at 7%, not zero, so no state looks falsely affordable. The infographic pairs map and histogram at equal weight on a letter-size grid. The dashboard separates KPI cards from the map to prevent overlapping labels. The slide deck uses a dark background for projection and samples 5,000 scatter points at low alpha to show density. All statistics are computed from the NDCP file, not hard-coded values.

0.1.9 Ethical Considerations

Data changes: I derived annual cost (weekly median times 52) and the affordability ratio. Zero or missing MHI rows were excluded from ratio math. State views filter to 2018; trend charts aggregate by year. No source records were deleted.

Legal and regulatory context: NDCP is federal open data with no PII. The technical guide documents imputation and variable definitions. This project is research, not legal or policy advice.

Presentation risks: Maps can overstate uniformity within states. The "99% exceed 7%" figure is accurate for the filtered sample but needs the 2018 cutoff and heuristic threshold noted. The weak labor correlation could be misread as "cost does not matter." I labeled correlation strength in charts and avoided causal language in titles.

Filtering transparency: Missing price or income rows were dropped for ratio charts. That choice is noted in subtitles and footers. Imputed values were kept by default, assuming DOL imputation is acceptable for national storytelling.

Sourcing: Data came from the DOL NDCP Excel file and was checked against the published technical guide. Summary stats match independent code recalculation.

Acquisition: Public dataset only; no scraped or private data.

Mitigation: Source, year range, and metric labels on every chart. Footnotes on imputation and data vintage. In production, link to the NDCP download and note post-2018 conditions may differ.

0.1.10 Lessons Learned

Next time I would lock a shared outline and color palette before building any charts, and run the imputation sensitivity test upfront. I would also deploy the dashboard as a hosted app rather than a static notebook export.

What I enjoyed most was turning a 227-column spreadsheet into one affordability ratio that actually lands with a policymaker audience. Building the infographic and seeing the geographic story on a single page was the highlight.

Kyle Gotzman | DSC640 Term Project | NDCP, U.S. Dept. of Labor, 2008-2018